AI-Assisted TCAD Optimization of GAA Nanosheet FETs

Project concept, research flow, and comparison with three base papers

Working direction: Power–Performance–Area (PPA) oriented optimization of GAA nanosheet geometry using TCAD, an AI surrogate model, multi-objective optimization, and final TCAD validation.

Purpose of this document

This document organizes the proposed project into a clear research flow and compares it with three uploaded base papers. The comparison is intended to show what each paper already solves, what is different in the proposed project, and where novelty must be demonstrated rather than assumed.

1. Proposed Abstract

Abstract — Gate-all-around (GAA) nanosheet field-effect transistors are promising devices for advanced CMOS because their geometry provides flexibility in electrostatic control, current drive, and device footprint. However, the dependence of electrical behavior on nanosheet geometry is nonlinear, and exhaustive TCAD-based design-space exploration can be computationally expensive. This project proposes an AI-assisted TCAD methodology for exploring the power, performance, and area trade-offs of GAA nanosheet FETs. A controlled set of device geometries—initially focusing on parameters such as gate length, nanosheet width, and nanosheet thickness—will be evaluated using TCAD. Electrical quantities such as ON current, OFF current, threshold voltage, subthreshold swing, DIBL, and gate capacitance will be extracted from the simulations. These TCAD results will be used to train and evaluate a surrogate machine-learning model that rapidly predicts device behavior across the design space. The surrogate will then support multi-objective exploration to identify Pareto-optimal geometries under the selected PPA formulation and device-quality constraints. Selected candidate designs will be returned to TCAD for validation. The intended outcome is not merely an accurate ML predictor, but a TCAD-validated design-space methodology that explains the trade-offs between geometry, device characteristics, and PPA-oriented objectives.

Important scope note: The proposed work should not claim that AI + TCAD + GAA + multi-objective optimization is itself new. Xu et al. (2022) already demonstrate ML-based multi-objective optimization of stacked GAA nanosheet geometry and source/drain doping. The proposed differentiation therefore needs to come from the specific PPA-oriented formulation, selected design variables/constraints, evaluation methodology, and demonstrated TCAD-validated design insights.

2. Core Research Question

Primary question: Can an AI-assisted surrogate reduce the cost of exploring GAA nanosheet geometry while preserving enough accuracy to identify TCAD-valid designs with favorable power–performance–area trade-offs?

Secondary questions: (1) Which geometry parameters dominate the PPA trade-offs? (2) Which electrical/device metrics explain those trade-offs? (3) How accurately does the surrogate reproduce TCAD behavior? (4) How much simulation effort can be avoided? (5) Do the selected Pareto candidates remain good when re-simulated in TCAD?

3. Complete Project Flow

Stage 1 — Define the device and design space

Start with a controlled GAA nanosheet structure and choose a small number of geometry variables. A practical first set is L_g (gate length), W_{NS} (nanosheet width), and T_{NS} (nanosheet thickness). Process/material parameters should initially be held fixed so that the study does not become too large.

Stage 2 — Generate TCAD ground truth

Run a designed set of TCAD simulations over the geometry space. For each geometry, extract the electrical characteristics needed by the later stages. The TCAD results are the ground truth against which the AI model and optimized candidates are judged.

Stage 3 — Extract device metrics

Candidate outputs include I_{ON}, I_{OFF}, V_{TH}, SS, DIBL and capacitance-related quantities. These should be defined precisely before optimization. In particular, I_{OFF} should not automatically be called total power, and I_{ON} should not automatically be treated as speed.

Stage 4 — Train and validate the AI surrogate

Use geometry as input and TCAD-derived quantities as targets. The model should be evaluated on held-out geometries, not only on training data. Prediction error and physical plausibility matter because optimization will amplify model errors if the surrogate is unreliable.

Stage 5 — Formulate PPA objectives

Translate the device outputs into explicitly defined proxies or calculated quantities for power, performance, and area. Performance may be represented by an intrinsic delay/speed formulation rather than I_{ON} alone. Area at device level should be treated carefully as a footprint/density proxy unless a circuit/layout area is actually modeled.

Stage 6 — Multi-objective optimization

Search the surrogate-defined design space for trade-offs instead of forcing a single universal optimum. The expected output is a Pareto frontier: designs for which improving one objective would require sacrificing another.

Stage 7 — TCAD validation

Select representative Pareto candidates and re-run the original TCAD simulations. Compare predicted and simulated values and report the error. This closes the loop between AI optimization and physical device simulation.

Input	Operation	Output
L_g, W_{NS}, T_{NS}	Geometry definition + DOE/design-space sampling	TCAD device structures
TCAD structures	Electrical simulation	I_D-V_G, I_D-V_D, fields/capacitance as available
TCAD curves	Metric extraction	I_{ON}, I_{OFF}, V_{TH}, SS, DIBL, C_G, etc.
Geometry + TCAD labels	ML surrogate training	Fast predicted device metrics
Predicted metrics	PPA formulation + constraints	Power/performance/area objective values
Surrogate	Multi-objective search	Pareto-optimal candidate geometries
Selected candidates	TCAD re-simulation	Validated final designs + error/trade-off analysis

4. Three Base Papers: What Each One Actually Does

Base Paper 1 — Xu et al. (2022)

Title: “A Machine Learning Approach for Optimization of Channel Geometry and Source/Drain Doping Profile of Stacked Nanosheet Transistors.” The paper proposes an ML-based multi-objective workflow for sub-3-nm, three-layer stacked GAA nanosheet transistors. An ANN learns the compact I–V relationship from 3-D TCAD simulations, and multi-objective optimization is performed over threshold swing, ON/OFF ratio, and ON-state current using an adaptive weighted-sum approach. The paper therefore establishes a very close precedent for TCAD → ANN → multi-objective optimization in GAA nanosheets.

Why it matters to our project: This is the most important paper for defining novelty. It means our contribution cannot simply be “use ML to optimize GAA nanosheets.” Our research question must clearly differ in objective formulation, design variables, constraints, evaluation, or demonstrated engineering outcome.

Base Paper 2 — Patel et al. (2025)

Title: “Machine Learning Augmented TCAD Assessment of Corner Radii in Nanosheet FET.” The study uses Sentaurus TCAD for a single-stack three-sheet nanosheet FET and investigates the effect of corner radius. It uses XGBoost to create a local fast simulator and uses stacked autoencoders and InfoGAN-based data augmentation to improve generalization. The study examines quantities including I_{ON}, I_{OFF}, C_{gg}, SS, I_{ON}/C_{gg}, f_T, and f_{max}.

Why it matters to our project: It demonstrates that ML can be used as a computationally efficient extension of TCAD and that geometry details such as corner rounding can materially affect current, leakage, capacitance and RF performance. Our project can learn from its surrogate-validation philosophy without making corner radius the main variable initially.

Base Paper 3 — Huang et al. (2021)

Title: “TCAD-Based Assessment of the Lateral GAA Nanosheet Transistor for Future CMOS.” This work is primarily a physical TCAD assessment rather than an AI optimization study. It studies lateral GAA nanosheet behavior at a projected G40M16 node and emphasizes the consequences of nanosheet width, thickness, parasitic capacitance, short-channel effects, series resistance and device footprint/density.

Why it matters to our project: This paper provides a strong physical basis for why geometry should be optimized rather than simply maximized. It shows that narrower and wider nanosheets create competing effects involving capacitance, electrostatics, resistance and density. It is therefore useful for defining the engineering interpretation of the Pareto trade-offs.

5. Direct Comparison

Aspect	Xu 2022	Patel 2025	Huang 2021	Proposed project
Main purpose	ML-based MOO of stacked GAA NSFET geometry + S/D doping	ML-augmented TCAD for corner-radius analysis	Physical TCAD assessment of lateral GAA geometry and footprint	AI-assisted PPA-oriented GAA geometry exploration
TCAD	3-D TCAD	Sentaurus TCAD	3-D numerical device simulation	TCAD as ground truth + final validation
AI	ANN	XGBoost + SAE/InfoGAN augmentation	None as core method	Surrogate ML model; exact model to be selected after baseline study
Main variables	Channel geometry + S/D doping profile	Corner radius / corner variation	Sheet width + thickness and related physical effects	Initially Lg, WNS, TNS; later expansion only if justified
Main objectives	SS, log(ION/IOFF), ION	DC/RF performance metrics and fast prediction	Current, capacitance, SCE, resistance, footprint/density	Power + performance + area, with device-quality constraints
Optimization	Multi-objective Pareto/weighted-sum approach	Design assessment / surrogate acceleration	Parameter trade-off analysis	Multi-objective Pareto search using surrogate predictions
Final validation	Optimized design window against TCAD/electrical targets	ML predictions checked against TCAD data	TCAD physical assessment	Re-simulate selected Pareto candidates in TCAD
Key lesson for us	Avoid claiming ML+TCAD+MOO as novelty	ML can reduce simulation burden	PPA trade-offs must be physically interpreted	Combine these lessons around a clearly defined PPA methodology

6. Where the Proposed Project Fits

The three papers form a useful progression. **Huang** gives the physical understanding of GAA geometry trade-offs. **Xu** shows that ML can learn TCAD behavior and support multi-objective optimization of nanosheet devices. **Patel** shows a more recent example of using ML to make TCAD-based analysis faster and more robust for a specific geometric effect.

The proposed project attempts to connect these lessons around a PPA-oriented design question: **which GAA nanosheet geometries are attractive when power, performance and area are considered simultaneously, and can AI reduce the number of expensive TCAD evaluations needed to find them?**

Conceptual differentiation

Not the contribution	Potential contribution
“We used ANN on TCAD data.”	A validated surrogate-based exploration methodology with explicit accuracy and simulation-saving evaluation.
“We performed multi-objective optimization.”	PPA-oriented objectives/constraints tied to device physics and footprint, with a clearly reported Pareto frontier.
“We found the best geometry.”	We characterize a family of trade-off solutions and explain why different geometries favor different objectives.
“AI predicts TCAD.”	AI proposes candidate designs, then TCAD independently validates the selected candidates.

7. Expected Results

Result A — Device-level understanding: plots showing how the selected geometry parameters change I_{ON}, I_{OFF}, V_{TH}, SS, DIBL and capacitance-related behavior.

Result B — Surrogate quality: held-out prediction accuracy, error distributions and—where appropriate—comparison of different baseline ML models.

Result C — PPA trade-off map: a Pareto frontier or equivalent multi-objective design-space visualization showing candidate geometries for different priorities.

Result D — TCAD validation: re-simulated Pareto candidates demonstrating how closely the AI-selected designs match the original TCAD predictions.

Result E — Computational benefit: an estimate of how many TCAD evaluations are avoided compared with exhaustive or baseline search. This should be measured rather than assumed.

8. Critical Definitions to Lock Before Implementation

The uploaded research report emphasizes that "optimized" is multi-dimensional and that the metric definitions must be explicit. For this project, the following are design decisions rather than already-fixed facts:

Question	What must be decided
Power	Will you use leakage as a proxy, calculate a specific static/dynamic quantity, or include capacitance/current under a defined operating condition?
Performance	Will you use an intrinsic delay/speed metric derived from capacitance and ON current, or another clearly justified device-level metric?
Area	What physical footprint/density proxy can be extracted consistently from your device geometry?
Constraints	Which VTH, SS, DIBL, leakage or other device-quality limits must every Pareto candidate satisfy?
AI evaluation	What error threshold is acceptable, and how will you prove that optimization conclusions remain valid under surrogate error?

9. Recommended Project Boundary

To keep the project scientifically defensible, the first version should stay at one device level and use a small geometry space. Do not simultaneously vary geometry, doping, material, strain, work function, corner radius, sheet count and process conditions. Once the core surrogate → optimization → TCAD-validation loop works, one additional variable can be introduced only if it answers a specific research question.

Suggested first boundary: three geometry variables → TCAD electrical outputs → one or two ML baselines → explicitly defined PPA objectives → Pareto search → TCAD validation.

10. Key Takeaway

The strongest version of this project is not "AI replaces TCAD." It is "AI helps navigate an expensive TCAD design space, while TCAD remains the physical ground truth." Your existing TCAD experience should therefore remain central: the AI layer should accelerate and organize the design exploration, while the final engineering conclusions should be explained using device physics.

References / Base Sources

1. H. Xu et al., "A Machine Learning Approach for Optimization of Channel Geometry and Source/Drain Doping Profile of Stacked Nanosheet Transistors," IEEE Transactions on Electron Devices, vol. 69, no. 7, 2022. DOI: 10.1109/TED.2022.3175708.

2. J. Patel et al., "Machine learning augmented TCAD assessment of corner radii in nanosheet FET," Solid-State Electronics, vol. 227, 109114, 2025. DOI: 10.1016/j.sse.2025.109114.

3. Y.-C. Huang, M.-H. Chiang, S.-J. Wang, and J. G. Fossum, "TCAD-Based Assessment of the Lateral GAA Nanosheet Transistor for Future CMOS," IEEE Transactions on Electron Devices, vol. 68, no. 12, 2021.

4. Uploaded supporting report: "AI-Driven Optimization Analysis for VLSI / TCAD Designs," August 2026.